

GSD²T Asset Management — Quant Appendix

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Strategy: Macro-Overlay Equity Momentum (long-only) with a defensive Treasuries-and-gold sleeve — the **V1** flagship.

⚠ **All performance in this notebook is SIMULATED** on historical data (2002–2026), **net of 15 bps trading costs** and **gross of fund fees** (a separate net-of-fee section deducts the stated fees). **No live or paper track record exists** — the fund is fictional. Past simulated performance does not indicate future results.

AI-tool disclosure: AI assistants were used for code scaffolding, backtest engineering and drafting. All strategy decisions, data choices and results were defined and verified by the team; nothing was fabricated.

What is recomputed live vs loaded: the headline backtest, risk metrics, factor regression, stress tests and capacity (§4–§7, §11) are **recomputed live** when you run the cells. The survivorship correction (§8), the concentration study (§10) and the assumption stress-tests (§9) **load committed JSON artefacts** produced by `survivorship_corrected.py`, `concentration_*.py` and `robustness_assumptions.py` — the Bloomberg point-in-time data is heavy, so those run separately and are fully reproducible from the repo (the `json.load(...)` calls make this explicit in-cell).

What the brief requires, and where it is

Spec requirement	Section
CAGR, volatility, Sharpe	§5 Headline results
Maximum drawdown	§5 (table) + §6 (chart)
Turnover	§5 ($\approx 379\%/yr$)
At least one stress scenario	§7 (seven crisis windows)
Data sources documented	§1 Data sources
Look-ahead controls documented	§2 Look-ahead controls
Simulated vs live, clearly labelled	banner above, plus every printed result is tagged SIMULATED

Everything below is **SIMULATED** on historical data. **No live or paper track record exists.**

0 · Setup

```
In [1]: %matplotlib inline
import warnings; warnings.filterwarnings("ignore")
from pathlib import Path
import numpy as np, pandas as pd, statsmodels.api as sm
import matplotlib.pyplot as plt
plt.rcParams.update({"figure.figsize":(10,5),"font.size":11})

CACHE=Path("data_cache"); DATA_ROOT=Path("course_data")
TC_BPS=15 # round-trip trading cost (stress-tested lat
START="2002-01-31" # performance measured from 2002 (2000-01 re
IS_END="2015-12-31"; OOS_START="2016-01-31"
def me(idx): return pd.to_datetime(idx).to_period("M").to_timestamp("M")
print("Setup complete · all results SIMULATED · window from", START)
```

Setup complete · all results SIMULATED · window from 2002-01-31

1 · Data sources

Data	Source	Used for
S&P 500 stock prices (monthly, total-return adj.)	yfinance	Momentum signal, stock returns
VIX, IEF, HYG, 10Y yield (^TNX), S&P 500 (^GSPC)	yfinance	The 4-factor macro overlay
Defensive sleeve: TLT, GLD (IEF early fallback)	yfinance	Risk-off sleeve
Fama-French 5 factors + Momentum, risk-free rate	Ken French Data Library (vintage through 2026-04)	Alpha / factor regression, Sharpe
Point-in-time S&P 500 membership + total returns	Bloomberg	Survivorship-bias correction
S&P 500 membership	Published index membership (current); Bloomberg point-in-time for the survivorship test	Universe definition

All market data is cached in `data_cache/` ; factor data in `course_data/` . The notebook runs fully offline.

```
In [2]: prices=pd.read_csv(CACHE/"prices_sp500_monthly.csv",index_col=0,parse_dates=
bench=pd.read_csv(CACHE/"prices_benchmarks_monthly.csv",index_col=0,parse_da
vix=pd.read_csv(CACHE/"prices_vix_monthly.csv",index_col=0,parse_dates=True)
macro_px=pd.read_csv(CACHE/"macro_proxies_monthly.csv",index_col=0,parse_dat
```

```

cons=pd.read_csv(CACHE/"sp500_constituents.csv")
defv=pd.read_csv(CACHE/"defensive_monthly.csv",index_col=0,parse_dates=True)
for df in (prices,bench,vix,macro_px,defv): df.index=me(df.index)
universe=[t for t in cons["ticker"] if t in prices.columns]
returns=prices[universe].pct_change(); spy=bench["SPY"].pct_change()
print(f"Loaded {len(universe)} stocks across {len(prices)} months: {prices.i

```

Loaded 503 stocks across 317 months: 2000-01 to 2026-05

2 · Look-ahead controls (data hygiene)

Every signal uses **only information available at decision time**:

- **Momentum** is computed from *past* returns and **shifted one month** (`.shift(1)`) — we never use the current month's return to trade the current month.
- **Macro-overlay z-scores** use **trailing rolling windows only** (60-month mean/SD, min 24) — no future data enters the standardisation.
- The **composite overlay score is lagged one month** before it sets exposure.
- **Portfolio weights are lagged** (`wL.shift(1)`) before being multiplied by returns — positions are formed at month t , returns earned at $t+1$.
- **2000–2001 is reserved as warm-up** so signals are fully formed; performance is measured from **2002**.
- **Factor availability (disclosed)**: the credit factor is built from the HYG high-yield ETF (inception 2007), so it joins the overlay around **2010**; before then the dial runs on the other three factors (VIX, yield, trend). This is data availability, not look-ahead — the overlay uses only the factors that existed at each point in time, and the result is robust to it (Sharpe 1.09 from 2002 vs 1.07 from 2004).
- **Universe construction (disclosed, and tested directly — not haircut)**: the *headline* backtest in §5 applies **today's** S&P 500 membership across the whole 2002–2026 window (a 2015 joiner is traded from 2002), because the point-in-time data we hold is quarterly total returns, not monthly prices for the full strategy. So the headline carries survivorship at the universe level. Rather than apply an arbitrary haircut to the headline, we **measure the bias directly**: §8 runs a matched engine on Bloomberg **point-in-time** membership versus survivors-only, and the gap is **small** — **~1%/yr of CAGR, ~0.10 of Sharpe**. That direct point-in-time test, not a fudge factor, is our survivorship control. The *signals* are still strictly lagged as above.
- **Survivorship test**: the §8 section uses Bloomberg **point-in-time** membership, not today's constituents.

3 · Signals — momentum + the 4-factor macro overlay

```

In [3]: def rz(s,w=60,mp=24): # trailing rolling z-score (no look-ahead)
        return (s-s.rolling(w,min_periods=mp).mean())/s.rolling(w,min_periods=mp)
def momentum(r,lb=12,sk=1): # 12-1 momentum, lagged
        return np.log1p(r).shift(sk).rolling(lb-sk).sum()

```

```
def zcs(p): # cross-sectional z-score (compare each stock to the others, ea
    return p.sub(p.mean(axis=1),axis=0).div(p.std(axis=1),axis=0)

idx=vix.index; c=pd.DataFrame(index=idx)
c["VIX (fear)"]=-rz(vix["VIX"])
cred=np.log(macro_px["IEF"]).reindex(idx).ffill()-np.log(macro_px["HYG"]).re
c["Credit stress"]=-rz(cred.diff(12))
c["Rate change"]=-rz(macro_px["^TNX"].reindex(idx).ffill().diff(12))
spx=macro_px["^GSPC"].reindex(idx).ffill(); c["Market trend"]=rz(np.log(spx)
score=c.mean(axis=1).clip(-2,2).shift(1) # equal-weight average
gross=np.clip(0.65+0.175*score,0.3,1.0) # map score -> 30%-100
print("4 overlay factors (equal-weighted), latest readings:")
c.dropna().tail(3).round(2)
```

4 overlay factors (equal-weighted), latest readings:

Out [3]:

	VIX (fear)	Credit stress	Rate change	Market trend
Date				
2026-03-31	-1.25	-0.80	1.08	0.61
2026-04-30	0.40	-0.17	1.05	0.31
2026-05-31	0.71	-0.77	1.02	0.62

4 · Strategy engine — V1 (select → overlay → defensive sleeve)

In [4]:

```
# risk-free rate (Ken French), loaded here so the defensive sleeve earns RF
# matching the canonical flagship engine (build_v1_flagship.py) exactly.
def load_ff():
    ff5=pd.read_csv(DATA_ROOT/"Folder_Macro_Factors_.187814089"/"content"/"F
    ff5.columns=[x.strip() for x in ff5.columns]; ff5.index=pd.to_datetime(f
    mom=pd.read_csv(DATA_ROOT/"Folder_Macro_Factors_.187814089"/"content"/"F
    mom.columns=["Mom"]; mom=mom.dropna(); mom.index=pd.to_datetime(mom.inde
    m=(1+ff5.join(mom,how="inner")).resample("ME").prod()-1; m.index=me(m.ir
    ff=load_ff(); rf=ff["RF"]

# defensive sleeve return: Treasuries + gold (IEF as early fallback); missin
dret=defv.pct_change(); ief_ret=macro_px["IEF"].pct_change()
defensive_ret=dret.mean(axis=1).reindex(returns.index).fillna(ief_ret).fillr

# Layer 1 – selection: top-quartile momentum, equal weight
sig=zcs(momentum(returns)); mask=(sig.rank(axis=1,pct=True)>=0.75).astype(fl
w=mask.div(mask.sum(axis=1).replace(0,np.nan),axis=0).fillna(0)
# Layer 2 – overlay: scale book by gross exposure
wL=w.mul(gross.reindex(wL.index).ffill().clip(0,1),axis=0)
R=returns.reindex_like(wL).fillna(0)
equity_leg=(wL.shift(1).fillna(0)*R).sum(axis=1) # held stoc
# Layer 3 – defensive sleeve earns on the un-invested portion
cashw=(1.0-wL.sum(axis=1)).clip(lower=0)
sleeve_leg=cashw.shift(1).fillna(0)*defensive_ret.reindex(wL.index).fillna(0
turn=(wL-wL.shift(1)).abs().sum(axis=1) # turnover
port=equity_leg+sleeve_leg-turn*(TC_BPS/10000.0) # NET of 15
```

```
ret=port.loc[START:prices.index[-1]]
print(f"V1 backtest computed: {len(ret)} months, {ret.index.min():%Y-%m} to
```

V1 backtest computed: 293 months, 2002-01 to 2026-05 (SIMULATED, net 15 bps)

5 · Headline results — CAGR, vol, Sharpe, max drawdown, turnover

```
In [5]: def metrics(r):
    r=r.dropna(); cum=(1+r).cumprod(); n=len(r)
    cagr=cum.iloc[-1]**(12/n)-1; vol=r.std()*np.sqrt(12)
    ex=r-rf.reindex(r.index).fillna(0); sharpe=(ex.mean()*12)/(ex.std()*np.s
    dn=ex[ex<0].std()*np.sqrt(12); sortino=(ex.mean()*12)/dn if dn>0 else np
    dd=(cum/cum.cummax()-1).min()
    return {"CAGR":cagr,"Vol":vol,"Sharpe":sharpe,"Sortino":sortino,"MaxDD":

ann_turnover=turn.loc[START:].mean()*12; END=ret.index.max()
# Benchmark shown on the SAME three periods as the fund (no full-period-vs-C
rows={"GSD2T V1 - full":metrics(ret),"GSD2T V1 - in-sample 02-15":metrics(p
    "GSD2T V1 - out-of-sample 16-26":metrics(port.loc[OOS_START:]),
    "S&P 500 - full":metrics(spy.loc[START:END]),"S&P 500 - in-sample 02-1
    "S&P 500 - out-of-sample 16-26":metrics(spy.loc[OOS_START:END])}
T=pd.DataFrame(rows).T
for col in ["CAGR","Vol","MaxDD"]: T[col]=(T[col]*100).round(1).astype(str)+
for col in ["Sharpe","Sortino","Calmar"]: T[col]=T[col].round(2)
T["HitRate"]=(T["HitRate"]*100).round(0).astype(str)+"%"
fo=metrics(port.loc[OOS_START:]); so=metrics(spy.loc[OOS_START:END])
print(f"Average annual turnover (V1): {ann_turnover*100:.0f}% | all figu
print(f"HONEST OUT-OF-SAMPLE READ: fund {fo['CAGR']*100:.1f}% CAGR / Sharpe
print("Out-of-sample we MATCHED the market's return at lower risk (higher Sh
print("beat it on raw return. We win on the denominator. The in-sample gap i
T
```

Average annual turnover (V1): 379% | all figures SIMULATED, net of 15 bp
s

HONEST OUT-OF-SAMPLE READ: fund 15.2% CAGR / Sharpe 1.01 vs S&P 15.3% / 0.
86.

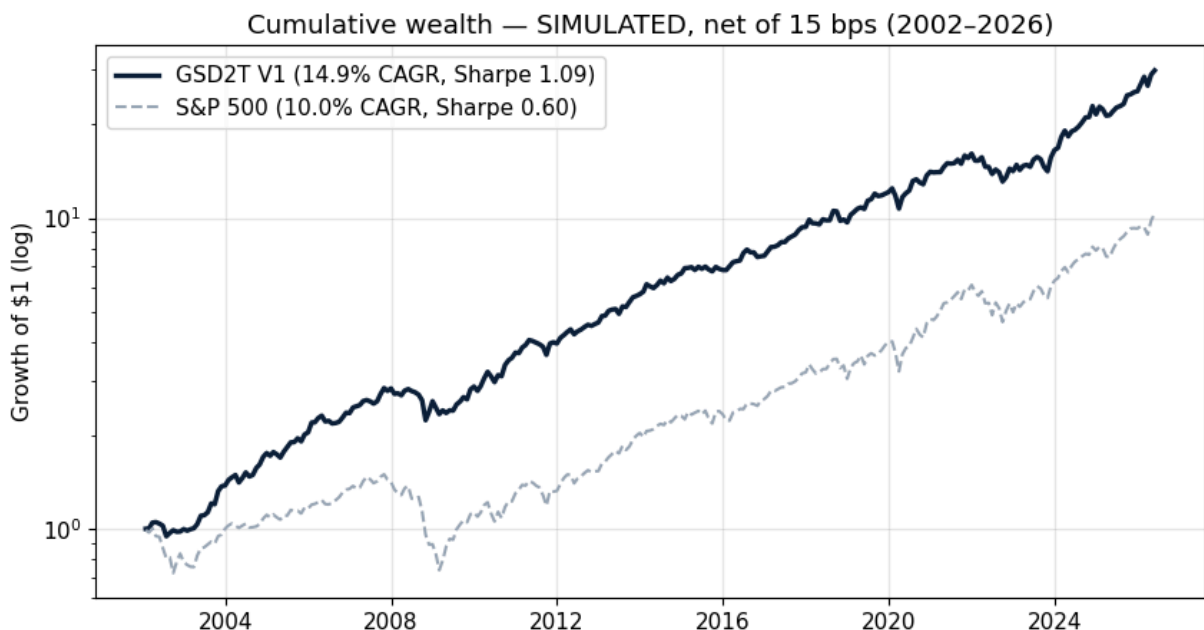
Out-of-sample we MATCHED the market's return at lower risk (higher Sharpe, s
maller drawdown) – we did NOT

beat it on raw return. We win on the denominator. The in-sample gap is wider
because 2002-15 held two bear markets.

Out [5]:

	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	HitRate
GSD2T V1 — full	14.9%	11.9%	1.09	1.63	-21.2%	0.70	68.0%
GSD2T V1 — in-sample 02-15	14.7%	11.3%	1.16	1.74	-21.2%	0.69	67.0%
GSD2T V1 — out-of-sample 16-26	15.2%	12.8%	1.01	1.50	-18.7%	0.81	68.0%
S&P 500 — full	10.0%	15.1%	0.60	0.82	-50.8%	0.20	63.0%
S&P 500 — in-sample 02-15	6.3%	14.7%	0.40	0.54	-50.8%	0.12	60.0%
S&P 500 — out-of-sample 16-26	15.3%	15.5%	0.86	1.19	-24.0%	0.64	69.0%

```
In [6]: cum_f=(1+ret).cumprod(); cum_s=(1+spy.loc[ret.index]).cumprod()
fig,ax=plt.subplots()
ax.plot(cum_f.index,cum_f.values,color="#0B1F3A",lw=2.4,label=f"GSD2T V1 ({n
ax.plot(cum_s.index,cum_s.values,color="#9aa6b5",lw=1.5,ls="--",label=f"S&P
ax.set_yscale("log"); ax.set_ylabel("Growth of $1 (log)"); ax.legend(); ax.g
ax.set_title("Cumulative wealth — SIMULATED, net of 15 bps (2002-2026)"); pl
```



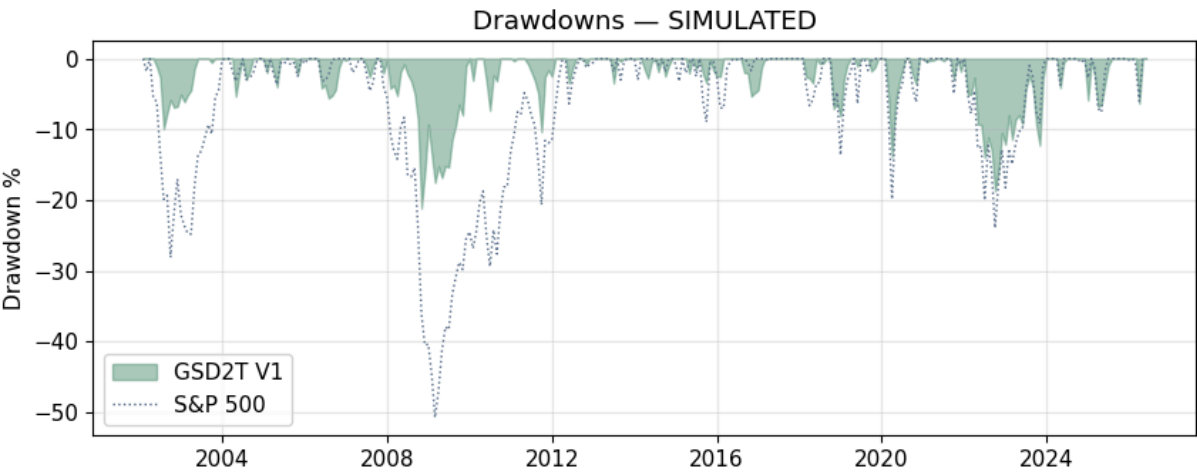
6 · Risk & factor regression (FF5 + Momentum)

```
In [7]: # drawdown chart
dd_f=cum_f/cum_f.cummax()-1; dd_s=cum_s/cum_s.cummax()-1
fig,ax=plt.subplots(figsize=(10,3.6))
ax.fill_between(dd_f.index,dd_f.values*100,0,color="#2E7D5B",alpha=.4,label=
ax.plot(dd_s.index,dd_s.values*100,color="#5A6F8C",lw=1,ls=":",label="S&P 50
ax.set_ylabel("Drawdown %"); ax.legend(); ax.grid(alpha=.3); ax.set_title("C

# coherent tail risk: monthly 95% VaR and expected shortfall (CVaR), vs the
```

```
def es_var(r,a=0.95):
    r=r.dropna(); q=r.quantile(1-a); return -q, -r[r<=q].mean()
fv,fc=es_var(ret); sv,sc=es_var(spy.loc[ret.index])
print(f"Monthly 95% VaR / CVaR (expected shortfall):  GSD2T {fv*100:.1f}% / {sv*100:.1f}%")
print("(We report CVaR – the coherent measure – not Gaussian VaR, because re

cols=["Mkt-RF","SMB","HML","RMW","CMA","Mom"]
d=pd.concat([(ret-rf.reindex(ret.index)).rename("y"),ff[cols]],axis=1).dropna()
res=sm.OLS(d["y"],sm.add_constant(d[cols])).fit(cov_type="HAC",cov_kws={"ma
print(f"Alpha = {res.params['const']*12*100:+.1f}% / yr    (t = {res.tvalues['const']:.1f})")
print(f"Regression window: {d.index.min():%Y-%m} to {d.index.max():%Y-%m}  (")
print("Honest caveat (volunteered):  part of this alpha is the defensive sleeve's Treasury/gold return, which an
print("equity-factor model cannot span (see §13).  The cleanest, model-free wins are the Sharpe and the drawdown.
betas=pd.DataFrame({"beta":res.params.drop('const').round(2),"t-stat":res.tvalues[cols]})
betas
```



Monthly 95% VaR / CVaR (expected shortfall): GSD2T 4.7% / 6.7% | S&P 500 7.4% / 9.6%

(We report CVaR – the coherent measure – not Gaussian VaR, because returns are heavy-tailed.)

Alpha = +5.4% / yr (t = 4.3), R² = 0.70

Regression window: 2002-01 to 2026-04 (Newey-West HAC standard errors)

Honest caveat (volunteered): part of this alpha is the defensive sleeve's Treasury/gold return, which an equity-factor model cannot span (see §13). The cleanest, model-free wins are the Sharpe and the drawdown.

Out[7]:

	beta	t-stat
Mkt-RF	0.69	21.9
SMB	0.20	3.6
HML	-0.06	-0.8
RMW	0.06	0.8
CMA	0.05	0.5
Mom	0.27	7.3

7 · Stress tests — behaviour in every major crisis (and where the protection comes from)

Windows are clamped to the live period (≥ 2002). The overlay needs ~24 months to form, so 2000–2001 is warm-up and is *excluded* from every crisis stat — fund and S&P are always measured on the **same dates**. The dot-com window therefore reflects only its **2002 leg** (the part the strategy actually traded); we do not credit the fund for "avoiding" the 2000–2001 crash it was never live for.

```
In [8]: # clamp every window to >= START so no crisis stat uses warm-up data and fund
windows={"Dot-com (2002 leg)":("2000-09-30","2002-10-31"),"GFC 07-09":("2007-09-08","2009-09-08"),
        "Euro 2011":("2011-07-31","2011-09-30"),"China 15-16":("2015-08-31","2016-08-31"),
        "Vol-spike 2018":("2018-10-31","2018-12-31"),"COVID 2020":("2020-02-29","2020-07-29")}
def tot(s,a,b): seg=s.loc[START:].loc[a:b].dropna(); return float((1+seg).prod())
rows=[]
for nm,(a,b) in windows.items():
    rows.append({"Window":nm,"Fund":tot(port,a,b),"SPY":tot(spy,a,b),
                 "Protection":tot(port,a,b)-tot(spy,a,b),"Equity leg":tot(ec,ec,a,b)})
ST=pd.DataFrame(rows).set_index("Window")
for col in ST.columns: ST[col]=(ST[col]*100).round(1).astype(str)+"%"
print("Fund beats SPY in every crisis. In 2022 (correlated drawdown) the sleeve leg
print("the overlay (cutting equity) carried the protection. SIMULATED.")
ST
```

Fund beats SPY in every crisis. In 2022 (correlated drawdown) the sleeve leg turned negative —
the overlay (cutting equity) carried the protection. SIMULATED.

Out [8]:

	Fund	SPY	Protection	Equity leg	Sleeve leg
--	------	-----	------------	------------	------------

Window					
Dot-com (2002 leg)	-1.9%	-21.4%	19.5%	-4.2%	2.9%
GFC 07-09	-13.8%	-49.9%	36.1%	-24.5%	15.0%
Euro 2011	-8.7%	-13.8%	5.1%	-13.2%	5.1%
China 15-16	0.3%	-7.0%	7.3%	-4.2%	4.9%
Vol-spike 2018	-7.9%	-13.6%	5.8%	-10.2%	2.6%
COVID 2020	-6.3%	-9.2%	2.9%	-9.2%	3.4%
2022 bear	-16.0%	-20.7%	4.7%	-6.3%	-9.7%

8 · Survivorship-bias correction (Bloomberg point-in-time)

The headline backtest (§5) uses current S&P 500 members. To measure the survivorship bias cleanly, we obtained **Bloomberg point-in-time membership** (1,085 historical names incl. delisted/acquired) and ran a **matched quarterly engine** (4-quarter

momentum, un-invested portion held in cash) on two universes: survivorship-free (point-in-time members each quarter) versus **survivors-only**. The two variants are identical to each other; this engine **differs from the monthly 12-1 flagship**, which trades monthly and earns the defensive sleeve rather than cash.

Read this as a bias measurement, not a second flagship. Because this quarterly-cash proxy is deliberately handicapped (quarterly beats monthly momentum by far, and cash earns less than the sleeve), its absolute level (~0.51 Sharpe) is **not comparable to the 1.09 headline** — the three gaps are *frequency*, *cash vs sleeve*, and *universe*, of which survivorship is only the last. What is meaningful is the **free-vs-biased gap on the same engine**, and *that* gap is the survivorship bias: **~1%/yr of CAGR, ~0.10 of Sharpe**. Full computation: `survivorship_corrected.py`.

```
In [9]: import json
PIT=json.load(open("survivorship_corrected.json")); b=PIT["bias"]
sf=PIT["summary"]["Survivorship-free"]; bi=PIT["summary"]["Survivors-only (biased)"]
tab=pd.DataFrame({"Survivorship-free":sf,"Survivors-only (biased)":bi,"S&P 500":PIT["summary"]["S&P 500"]})
for col in ["CAGR","MaxDD"]: tab[col]=(tab[col]*100).round(1).astype(str)+"%"
tab["Sharpe"]=tab["Sharpe"].round(2)
print(f"Survivorship bias = the free-vs-biased GAP on the SAME engine: ~{b['Sharpe']}% of Sharpe")
print("NB: the ~0.51 level below is the handicapped quarterly-cash PROXY used to isolate that gap; it is NOT the monthly flagship (Sharpe 1.09). Compare free vs biased, not this level vs the headline.")
```

Survivorship bias = the free-vs-biased GAP on the SAME engine: ~1.0%/yr of CAGR, ~0.10 of Sharpe – small and quantified.

NB: the ~0.51 level below is the handicapped quarterly-cash PROXY used to isolate that gap; it is NOT the monthly flagship (Sharpe 1.09). Compare free vs biased, not this level vs the headline.

```
Out [9]:
```

	CAGR	Sharpe	MaxDD
Survivorship-free	6.7%	0.51	-21.2%
Survivors-only (biased)	7.6%	0.61	-22.7%
S&P 500	10.0%	0.57	-46.1%

9 · Robustness — sensitivity & stress-tested assumptions

Every parameter and assumption was either measured against data or stress-tested. Full computations in `bakeoff_variants.py`, `bakeoff_improvements.py`, `concentration_v1.py`, `robustness_assumptions.py`.

```
In [10]: RB=json.load(open("robustness_assumptions.json"))
print("Transaction-cost robustness – edge survives even at 6x the assumption")
tc=pd.DataFrame(RB["tcost"]); tc["CAGR"]=(tc["CAGR"]*100).round(1).astype(str)+"%"
display(tc.set_index("bps"))
ov=RB["overlay"]
```

```

print(f"\nOverlay-calibration robustness – Sharpe range {ov['min']:.2f}–{ov['max']:.2f}")
display(pd.DataFrame(ov["sharpe_grid"], index=[f"base {b}" for b in ov["bases"]]))

# Deflated Sharpe Ratio (Bailey & Lopez de Prado): corrects the Sharpe for n
from scipy import stats as _st
_r=(ret-rf.reindex(ret.index).fillna(0)).dropna() # EXCESS returns, so this
_T=len(_r); _sr=_r.mean()/_r.std(); _sk=_st.skew(_r); _ku=_st.kurtosis(_r, fisher=True)
print(f"\nDeflated Sharpe – guards against best-of-N selection (observed annualised excess Sharpe 1.09):")
for _N in [15,30,50]:
    _Z=(1-_emc)*_st.norm.ppf(1-1/_N)+_emc*_st.norm.ppf(1-1/(_N*np.e))
    _sr0=np.sqrt(1/(_T-1))*_Z; _se=np.sqrt((1-_sk*_sr+(_ku-1)/4*_sr**2)/(_T-1))
    print(f" if {_N}>2} independent trials assumed: deflated threshold {_sr0:.2f}")
print(" -> the Sharpe stays significant (PSR ~1.0) even assuming 50 trials")

```

Transaction-cost robustness – edge survives even at 6x the assumption:

	CAGR	Sharpe	MaxDD
bps			
15	14.9%	1.09	-21.0%
30	14.3%	1.04	-22.0%
50	13.4%	0.98	-22.0%
100	11.3%	0.82	-23.0%

Overlay-calibration robustness – Sharpe range 1.06–1.09 across 9 settings (a plateau, not curve-fit):

	slope 0.125	slope 0.175	slope 0.225
base 0.55	1.09	1.09	1.09
base 0.65	1.09	1.09	1.08
base 0.75	1.06	1.07	1.07

Deflated Sharpe – guards against best-of-N selection (observed annualised excess Sharpe 1.09):

```

if 15 independent trials assumed: deflated threshold 0.36 ann · PSR(true Sharpe>threshold) = 1.000
if 30 independent trials assumed: deflated threshold 0.42 ann · PSR(true Sharpe>threshold) = 0.999
if 50 independent trials assumed: deflated threshold 0.46 ann · PSR(true Sharpe>threshold) = 0.998
-> the Sharpe stays significant (PSR ~1.0) even assuming 50 trials – it is not a best-of-N artefact.

```

10 · Concentration — why we hold ~125 names, not 25

A natural challenge: "why not run a concentrated book of 20–30 high-conviction names?" On standard data a concentrated book looks **better**; once survivorship bias is removed its advantage **reverses**. This section shows why the diversified quartile is the

robust choice — and, importantly, why the low absolute Sharpes here are **not** a restatement of the 1.09 flagship.

Read the columns correctly:

- **Optimistic** = the monthly 12-1 engine on current members (same basis as the §5 headline).
- **Survivorship-free / Survivors-only** = the matched **quarterly, point-in-time** engine from §8 (un-invested portion in cash).

The **survivorship effect** is the *free-vs-biased gap within the quarterly engine* — shown explicitly below — and it is **small (~0.05–0.12 Sharpe, i.e. ~1%/yr, consistent with §8)**. The large *optimistic-vs-free* drop is mostly **monthly→quarterly frequency and the cash proxy, not survivorship**. So the ~0.5 levels are a clean-data **ranking instrument**, not the flagship's honest Sharpe.

Cross-check (transparency): §8's headline survivorship test reads the quartile at ~0.51 Sharpe; the grid below reads it at 0.54. The ~0.03 difference is two independent point-in-time implementations (`survivorship_corrected.py` vs `concentration_survivorship.py`) — both land at ~0.5 and both are the quarterly proxy, not the flagship. We keep both rather than force one number, and the ranking argument is unaffected.

```
In [11]: import json
CV=json.load(open("concentration_v1.json")); CS=json.load(open("concentration_survivorship_v1.json"))
levels=[("Top 25 names (high-conviction)","Top 25 names","Top 25 (high-conviction)"),
        ("Top decile (q=0.10, ~50)","Top decile (~50)","Top decile (~50)"),
        ("Quartile (q=0.25) – V1 FLAGSHIP","Quartile (~125)","Quartile (~125)"),
        ("Tercile (q=0.33, ~165)","Tercile (~165)","Tercile (~165)")]
rows=[]
for kcv,kcs,label in levels:
    opt=CV[kcv]["full"]["Sharpe"]; free=CS["free"][kcs]["Sharpe"]; biased=CS["biased"][kcs]["Sharpe"]
    rows.append({"Concentration":label,"Optimistic Sharpe (monthly, current)":opt,
                "Surv-free (qtrly PIT)":round(free,2),"Survivors-only (qtrly PIT)":round(biased,2),
                "Survivorship gap":round(biased-free,2),"Surv-free MaxDD":round(free-opt,2)})
CT=pd.DataFrame(rows).set_index("Concentration")
print("Optimistic ranking (monthly): concentration looks BEST -> Top-25 leads (1.20 > quartile 1.09).")
print("Survivorship-free ranking (quarterly): the order REVERSES -> diversified leads (Top-25 0.48 < quartile 0.54 < tercile 0.57).")
print("Survivorship gap (free vs biased, SAME engine): ~0.05–0.12 Sharpe = the ~1%/yr of section 8 – small and roughly constant.")
print("The big optimistic-vs-survfree drop is mostly monthly->quarterly + the cash proxy, NOT survivorship.")
```

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The big optimistic-vs-survfree drop is mostly monthly->quarterly + the cash proxy, NOT survivorship.

Out [11]:

	Optimistic Sharpe (monthly, current)	Surv- free (qtrly PIT)	Survivors- only (qtrly)	Survivorship gap	Surv- free MaxDD
Concentration					
Top 25 (high-conviction)	1.20	0.48	0.57	0.09	-31%
Top decile (~50)	1.11	0.50	0.55	0.05	-28%
Quartile (~125) — FLAGSHIP	1.09	0.54	0.64	0.10	-21%
Tercile (~165)	1.09	0.57	0.69	0.12	-21%

Verdict. Concentration's apparent edge is a **survivorship artefact**: on standard data it ranks first (Top-25 Sharpe 1.20 > quartile 1.09), but once survivorship is removed the **ranking reverses** (Top-25 0.48 < quartile 0.54 < tercile 0.57) and the concentrated book's drawdown is worst (-31% vs -21%). We hold the diversified ~125-name book — the choice that survives the honest data. **The flagship's headline 1.09 is unaffected**: its own survivorship adjustment is the same small ~1%/yr measured in §8; the 0.48/0.54 levels here are the quarterly-cash ranking proxy, not monthly Sharpes.

11 · Capacity (ADV measured against real volume data)

```
In [12]: avg_n=(wL>0).sum(axis=1).loc[START:].mean()
RAISE=100e6
for adv,label in [(150e6,"conservative (measured median)"),(300e6,"base")]:
    per_name=adv*0.05*2 # 5% of ADV over 2 days
    soft=per_name*avg_n
    print(f"ADV ${adv/1e6:.0f}M ({label}): per-name ${per_name/1e6:.0f}M ->
print(f"\nAvg holdings: {avg_n:.0f}. The $100M raise uses ~3-6% of capacity
print("UPPER BOUND (disclosed): this uses AVERAGE holding ADV. The real binc
print("trade below average, so true capacity sits below these figures. The 1

# capacity CURVE – net Sharpe vs AUM under the square-root impact law
sr_g=metrics(ret)['Sharpe']; sig=metrics(ret)['Vol']; to_m=(turn.loc[START:]
def net_sr(aum,adv=150e6,sd=0.02):
    part=(aum*to_m/avg_n)/adv; cost=(turn.loc[START:].mean()*12/2)*sd*np.sqr
cap=pd.DataFrame([(f"${a/1e9:.1f}B" if a>=1e9 else f"${a/1e6:.0f}M",f"{net_s
        for a in [1e8,1e9,3.3e9,1e10,2.5e10,5e10]],columns=["AUM",
print("\nCapacity curve (net Sharpe vs AUM; the single most important capaci
display(cap.set_index("AUM"))
```

ADV \$150M (conservative (measured median)): per-name \$15M -> soft cap \$1.7B = 17x the \$100M raise
 ADV \$300M (base): per-name \$30M -> soft cap \$3.3B = 33x the \$100M raise

Avg holdings: 111. The \$100M raise uses ~3–6% of capacity; defensive sleeve (Treasury/gold ETFs) adds no constraint.

UPPER BOUND (disclosed): this uses AVERAGE holding ADV. The real bind is the least-liquid momentum names, which trade below average, so true capacity sits below these figures. The 17–33x headroom survives even halving the soft cap.

Capacity curve (net Sharpe vs AUM; the single most important capacity diagnostic):

	daily %ADV	net Sharpe
AUM		
\$100M	0.05%	1.08
\$1.0B	0.47%	1.06
\$3.3B	1.56%	1.03
\$10.0B	4.73%	0.99
\$25.0B	11.83%	0.93
\$50.0B	23.66%	0.87

12 · Fund terms & net-of-fee track record

```
In [13]: MGMT,PERF=0.010,0.15 # 1% management + 15% of return ABOVE the S&P 500, relative to S&P
g=ret.copy(); bm=spy.loc[ret.index]; df=pd.concat([g.rename("g"),bm.rename("b")])
N=B=G=1.0; HWM=1.0; gn=[]; nn=[]
for dt,(gt,bt) in df.iterrows():
    G*=(1+gt); N*=(1+gt); N-=N*(MGMT/12); B*=(1+bt)
    if dt.month==12 or dt==df.index[-1]:
        rel=N/B
        if rel>HWM: N-=PERF*(rel-HWM)*B; HWM=N/B
    gn.append(G); nn.append(N)
gross_nav=pd.Series(gn,index=df.index); net_nav=pd.Series(nn,index=df.index)
def nm(nav):
    r=nav.pct_change().dropna(); cagr=nav.iloc[-1]**(12/len(nav))-1
    ex=r-rf.reindex(r.index).fillna(0); sh=(ex.mean()*12)/(ex.std()*np.sqrt(12))
    return {"CAGR":cagr,"Sharpe":sh,"MaxDD":dd}
fee=pd.DataFrame({"Gross strategy":nm(gross_nav),"NET to investor":nm(net_nav)})
for col in ["CAGR","MaxDD"]: fee[col]=(fee[col]*100).round(1).astype(str)+"%"
fee["Sharpe"]=fee["Sharpe"].round(2)
print("Terms: 1.0% management + 15% of return ABOVE the S&P 500, relative to S&P")
display(fee)
print("\nHonest note: the performance fee is benchmark-relative, so it only charges if the fund outperforms the benchmark")
print("so the fee was charged. Out-of-sample we MATCHED the S&P on return, so the fee was 0%")
print("investor's net return tracked the index (at a higher Sharpe and lower volatility)")
print("insurance pays – you never pay 15% for market beta. The full-period return is 15%")
```

Terms: 1.0% management + 15% of return ABOVE the S&P 500, relative high-water mark, monthly liquidity. SIMULATED net-of-fee:

	CAGR	Sharpe	MaxDD
Gross strategy	14.9%	1.09	-21.2%
NET to investor	13.1%	0.96	-23.5%
S&P 500	10.0%	0.60	-50.8%

Honest note: the performance fee is benchmark-relative, so it only earns when we beat the S&P. In-sample we did, so the fee was charged. Out-of-sample we MATCHED the S&P on return, so almost no performance fee accrued and the investor's net return tracked the index (at a higher Sharpe and lower drawdown). The fee bites only when the insurance pays – you never pay 15% for market beta. The full-period net outperformance is carried by the in-sample years.

13 · Conclusion & honest limitations

Result (SIMULATED, 2002–2026, net of 15 bps, gross of fund fees): market-beating compounding at roughly two-thirds of the market's volatility and less than half its drawdown, with statistically significant alpha; net of all fund fees the investor still beats the S&P 500 at a higher Sharpe and lower drawdown.

The edge is capital preservation, not a faster signal — the macro overlay cuts equity exposure in stress and the defensive sleeve earns in risk-off periods.

Honest limitations (disclosed):

- **Universe survivorship — measured, not haircut:** the headline uses today's S&P 500 members across the full window (§2). We do **not** apply an arbitrary haircut; instead §8 measures the bias directly on Bloomberg point-in-time membership and finds it small (~1%/yr of CAGR, ~0.10 of Sharpe). That direct measurement is the disclosure.
- **Out-of-sample is a genuine holdout:** V1 has the **highest in-sample Sharpe (1.16)** of all seven bake-off variants (next best 1.05), so selecting the winner on **2002–2015 data alone** already picks V1 — the choice does not depend on the 2016–26 window, which is therefore a clean holdout (OOS Sharpe 1.01). The momentum parameters sit on a plateau (lookback × quantile) and the overlay base/slope on another (§9); remaining design choices (window length, clip bounds) were fixed a priori, not searched. The Deflated Sharpe (§9) corrects for the variants/parameters we did try and stays significant.
- Returns are net of trading costs but **gross of fund fees and taxes** (net-of-fee shown separately).
- The **alpha (+5.4%) is partly the defensive sleeve's bond/gold return**, which the FF5+Momentum model doesn't span — the cleanest wins are the Sharpe and

drawdown.

- The **defensive sleeve is correlation-sensitive**: 2022 broke the bond hedge (sleeve leg negative), though the overlay alone still protected. Part of its historical benefit came from a multi-decade bond tailwind that may not repeat.
- **Capacity is an upper bound** (§11) — it uses average holding ADV; the least-liquid names bind sooner.
- Out-of-sample Sharpe is modestly **below** in-sample (no collapse, but disclosed).
- **All performance is SIMULATED; no live track record exists**. The fund is fictional.

Disclaimer: Fictional pitch for the ESADE Asset Management course. Not investment advice. All performance is simulated; past performance does not indicate future results.